Software Requirements Specification (SRS)

Collaborative Study Material & Quiz Platform

Field	Details
Project	Collaborative Study Material & Quiz Platform
Team No.	16
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Version	Date	Author	Change Summary	Approval
1.0	09-09-2026	Team 16	Initial complete SRS for academic project	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator	_____	_____	_____
Project Evaluator	_____	_____	_____

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Collaborative Study Material & Quiz Platform. The system is a web application intended to provide students with a centralized environment for creating, sharing, discovering, collaboratively improving, and evaluating study material and quizzes. This document establishes the expected system behaviour, interfaces, quality attributes, security controls, acceptance criteria, and traceability needed for implementation and evaluation.

1.2 Scope

The platform covers user authentication, study-material management, collaborative editing, comments and ratings, material search and organization, version history, quiz creation and attempts, multiple question types, timed quizzes, randomized questions, automatic evaluation, instant results, leaderboards, and quiz-performance history. Administrative functions include user and content management. The system is specified as a web application. The final technology stack is intentionally left open and will be selected during the design and implementation phases.

1.3 Audience

- Project developers and student team members
- QA/test engineers and evaluators
- Course instructors and project reviewers
- System/design documentation reviewers

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
API	Application Programming Interface
FR	Functional Requirement
NFR	Non-Functional Requirement
RTM	Requirements Traceability Matrix
Quiz Creator	Authorized user who creates and manages quizzes
Content Contributor	Authorized user who uploads or collaboratively edits study material
Version	A saved state of a study-material item retained for history/recovery
Leaderboard	Ranking of users based on configured quiz-performance criteria

2. Overall Description

2.1 Product Perspective

The Collaborative Study Material & Quiz Platform is a standalone web-based learning support system. It provides a common repository for study resources and an interactive quiz environment. Users access the platform through a web browser. The application communicates with its backend services and database to authenticate users, store content and quiz data, record collaboration activity, evaluate quiz attempts, and maintain user-performance history.

2.2 Major Product Functions

- Username/password authentication and role-based access.
- Upload, create, organize, search, filter, download, edit, and delete study material.
- Collaborative editing and version history for shared material.
- Ratings, comments, and discussions associated with study material.
- Quiz creation and management by authorized quiz creators.

- Multiple-choice, True/False, and fill-in-the-blank question types.
- Timed and randomized quiz attempts.
- Automatic evaluation and instant result display.
- Leaderboard and quiz-performance history.
- Administrative moderation and user/content management.

2.3 User Roles and Characteristics

Role	Characteristics	Main Responsibilities
Student/User	Primary learner; basic computer/web literacy expected.	Browse/search material, download resources, comment/rate, attempt quizzes, view results and history.
Content Contributor / Quiz Creator	User with additional content-authoring permissions.	Upload/create/edit study material, collaborate on shared resources, manage versions, create/manage quizzes.
Administrator	System-management role with elevated permissions.	Manage users, moderate content, manage reported material, oversee platform activity and configuration.

2.4 Operating Environment

- Client: modern web browser on desktop/laptop; responsive web access may be supported.
- Application server: web/application runtime selected during implementation.
- Database: relational or document-oriented database selected during design.
- Network: standard Internet or institutional network connection using HTTPS.
- File storage: server-side or managed storage for uploaded study documents.

2.5 Constraints

- The system must be implemented as a web application.
- Authentication must use username and password.
- The project is a four-member academic implementation, so functionality should remain achievable within the project schedule.
- Technology stack is not fixed at the SRS stage.
- Uploaded files must be subject to configured type/size restrictions.
- Role permissions must be enforced consistently across protected operations.

2.6 Assumptions and Dependencies

- Users have valid credentials and access to a supported browser.
- The platform has access to persistent storage for material, quizzes, and user records.
- Administrators are responsible for moderation and account management.
- The final deployment environment provides HTTPS and adequate storage.
- Internet/network connectivity is available for normal operation.

3. External Interface Requirements

3.1 User Interfaces

- Login page with username and password fields and clear authentication feedback.
- Dashboard showing available study material, subjects/topics, recent content, and quiz activity.
- Material page with metadata, preview/download controls, comments, ratings, and version information.
- Material editor for authorized contributors, including save/update and collaboration controls.
- Quiz builder with question-type selection, answer configuration, timing, and randomization settings.
- Quiz attempt interface with question navigation, timer where applicable, submission confirmation, and result display.
- Administrator interface for user and content moderation.

3.2 Hardware Interfaces

No specialized hardware is required. The system shall operate using standard user devices such as laptops or desktop computers with keyboard, pointing device, display, and network connectivity.

3.3 Software Interfaces

- Web browser interface for end users.
- Backend application services for authentication, content management, quiz processing, and reporting.
- Database interface for persistent user, content, quiz, comment, rating, version, and attempt records.
- File-storage interface for uploaded documents.

3.4 Communications

- All authenticated and sensitive communication shall use HTTPS/TLS.
- Client-server communication shall use structured request/response formats.
- The system shall provide appropriate error responses when a request cannot be completed.
- Session handling shall prevent unauthorized reuse of authenticated sessions.

4. System Features (Detailed Functional Requirements)

Each functional requirement below is stated using the 'shall' form and includes a priority, stakeholder/source, acceptance criterion, test-case reference, and dependencies. IDs follow the CSQP-F-### convention.

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case	Comments / Dependencies
CSQP-F-001	The system shall authenticate users using a unique username and password before granting access to protected functions.	Functional	High	Student/User; Admin	Valid credentials create an authenticated session; invalid credentials are rejected. Test: TC-AUTH-01	User account store
CSQP-F-002	The system shall enforce role-based permissions for Student/User, Content Contributor / Quiz Creator, and Administrator roles.	Functional	High	All stakeholders	A user can perform only operations permitted by the assigned role. Test: TC-AUTH-02	Role mapping required

CSQP-F-003	The system shall allow authorized contributors to upload study notes and PDF documents with configured file-size and file-type validation.	Functional	High	Content Contributor	A valid file uploads successfully; invalid type/size is rejected with a clear message. Test: TC-MAT-01	File storage
CSQP-F-004	The system shall allow authorized contributors to create study notes directly within the platform.	Functional	High	Content Contributor	A note can be created, saved, and displayed in the selected subject/topic. Test: TC-MAT-02	Editor required
CSQP-F-005	The system shall organize study material by subject and topic and allow users to browse the hierarchy.	Functional	Medium	Student/User	Selecting a subject/ topic displays its associated material. Test: TC-MAT-03	Material metadata
CSQP-F-006	The system shall provide search and filtering facilities for study material using supported metadata such as title, subject, topic, and contributor.	Functional	High	Student/User	Search/filter returns matching material and excludes unrelated records under the selected criteria. Test: TC-MAT-04	Search index/query
CSQP-F-007	The system shall allow users with appropriate permissions to download available study material.	Functional	Medium	Student/User	Selecting download returns the stored file or generated note document successfully. Test: TC-MAT-05	File storage
CSQP-F-008	The system shall allow authorized contributors to edit or delete study material they are permitted to manage.	Functional	High	Content Contributor; Admin	An authorized update changes the material; deletion removes it from normal user listings. Test: TC-MAT-06	Authorization
CSQP-F-009	The system shall allow users to rate study material using the configured rating scale and prevent duplicate ratings by the same user for the same item.	Functional	Medium	Student/User	A user's rating is stored once and subsequent rating updates replace the previous value. Test: TC-MAT-07	User identity
CSQP-F-010	The system shall allow users to post comments and participate in discussions associated with study material.	Functional	Medium	Student/User	A permitted comment is stored and displayed with author and timestamp; unauthorized submissions are rejected. Test: TC-MAT-08	Moderation
CSQP-F-011	The system shall maintain version history for collaboratively edited study material and allow authorized users to view previous versions.	Functional	High	Content Contributor; Admin	Each saved collaborative revision has a distinct version and can be viewed by authorized users. Test: TC-MAT-09	Version store
CSQP-F-012	The system shall allow multiple authorized contributors to collaborate on shared study material through controlled edit/update operations.	Functional	High	Content Contributor	Authorized contributors can update shared material and changes are recorded against the corresponding version history. Test: TC-COL-01	Concurrency/ versioning

CSQP-F-013	The system shall allow authorized quiz creators to create quizzes containing multiple-choice, True/False, and fill-in-the-blank questions.	Functional	High	Quiz Creator	A quiz containing supported question types can be saved and published. Test: TC-QUIZ-01	Quiz database
CSQP-F-014	The system shall allow quiz creators to configure quiz duration and enable or disable question randomization.	Functional	High	Quiz Creator	Configured duration is applied during attempts and randomization changes question order when enabled. Test: TC-QUIZ-02	Quiz engine
CSQP-F-015	The system shall allow eligible users to start, navigate, and submit a quiz attempt.	Functional	High	Student/User	A user can answer available questions and submit once; submitted attempts are recorded. Test: TC-QUIZ-03	Quiz availability
CSQP-F-016	The system shall automatically evaluate objective quiz responses and calculate the user's score according to the quiz's configured marking rules.	Functional	High	Student/User; Quiz Creator	The calculated score matches the configured correct answers and marking rules. Test: TC-QUIZ-04	Evaluation engine
CSQP-F-017	The system shall display quiz results immediately after submission when the quiz is configured for instant results.	Functional	High	Student/User	After submission, the result page displays score and applicable performance details. Test: TC-QUIZ-05	Evaluation complete
CSQP-F-018	The system shall maintain a history of a user's completed quiz attempts, scores, and relevant timestamps.	Functional	Medium	Student/User	A completed attempt appears in the user's history with correct score and date/time. Test: TC-QUIZ-06	Attempt store
CSQP-F-019	The system shall provide a leaderboard for quizzes using the configured ranking criteria.	Functional	Medium	Student/User; Quiz Creator	Leaderboard displays ranked eligible attempts according to the configured criteria. Test: TC-QUIZ-07	Ranking logic
CSQP-F-020	The system shall allow administrators to manage user accounts, including viewing, enabling/disabling, and applying role changes where permitted.	Functional	High	Administrator	Admin changes are reflected in subsequent authorization checks. Test: TC-ADMIN-01	Admin authorization
CSQP-F-021	The system shall allow administrators to moderate study material, comments, and other user-generated content.	Functional	High	Administrator	Admin can remove or restrict flagged content and the change is reflected to normal users. Test: TC-ADMIN-02	Moderation rules
CSQP-F-022	The system shall record important user and administrative actions in an audit log without storing plaintext passwords.	Functional	High	Administrator; Security	Specified actions create timestamped audit events and password values are absent from logs. Test: TC-SEC-01	Audit storage

5. Non-Functional Requirements (Detailed)

The following measurable NFRs use IDs CSQP-NF-### and are tied to acceptance measurements.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
CSQP-NF-001	For normal load, 90% of standard page/API interactions shall complete within 3 seconds, excluding user-controlled file-transfer time.	Performance	High	90th percentile response time ≤ 3 seconds in production-like testing. Test: TC-PERF-01
CSQP-NF-002	The platform shall achieve at least 99.0% monthly availability excluding planned maintenance.	Reliability	High	Availability report shows ≥99.0% during the measurement period. Test: TC-REL-01
CSQP-NF-003	The user interface shall provide consistent navigation, readable feedback, and responsive layouts for supported desktop/laptop browser sizes.	Usability	Medium	Usability checklist passes for navigation, feedback, and layout at defined resolutions. Test: TC-UX-01
CSQP-NF-004	The system shall preserve data integrity so that saved material, quiz definitions, results, and user records are not lost or silently corrupted during normal operation.	Data Integrity	High	Create/update/retrieve tests return the same verified records and recovery tests preserve committed data. Test: TC-DATA-01
CSQP-NF-005	The platform shall support maintainable modular components and documented interfaces so that individual modules can be modified without requiring unrelated modules to be rewritten.	Maintainability	Medium	Architecture/code review confirms modular separation and documented interfaces. Test: TC-MAINT-01
CSQP-NF-006	The platform shall support the configured study-material and quiz workload without exceeding defined resource limits under normal academic usage.	Scalability	Medium	Load test meets response-time and resource thresholds for the agreed concurrent-user target. Test: TC-SCALE-01

5.1 Security Objectives

Objective	Description
SO-01 — Confidentiality	Protect user credentials, account information, private content controls, and other sensitive application data from unauthorized access or disclosure.
SO-02 — Integrity & Accountability	Prevent unauthorized modification of study material, quizzes, results, and administrative records while maintaining sufficient audit information to identify important actions.

5.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case
CSQP-SR-001	The system shall store passwords only as secure one-way password hashes using an approved password-hashing mechanism and shall never store plaintext passwords.	Security	High	Password database inspection and authentication tests show no plaintext password storage. Test: TC-SEC-02
CSQP-SR-002	The system shall use HTTPS/TLS for authenticated sessions and other sensitive client-server communication.	Security	High	Network inspection confirms sensitive traffic is protected by HTTPS/TLS. Test: TC-SEC-03
CSQP-SR-003	The system shall enforce server-side authorization checks for all protected study-material, quiz, administrative, and user-management operations.	Security	High	Direct unauthorized requests are rejected even when the UI is bypassed. Test: TC-SEC-04
CSQP-SR-004	The system shall validate and sanitize user-provided content and uploaded-file metadata to reduce risks from malicious input and unsafe file handling.	Security	High	Security test suite rejects unsafe inputs according to configured validation rules. Test: TC-SEC-05
CSQP-SR-005	The system shall terminate or invalidate authenticated sessions according to the configured session timeout/logout policy and shall prevent access to protected pages after logout.	Security	High	After logout or configured timeout, protected requests require re-authentication. Test: TC-SEC-06
CSQP-SR-006	The system shall prevent users from altering quiz scores or other authoritative result records through client-side manipulation.	Security	High	Modified client requests cannot change stored authoritative scores. Test: TC-SEC-07

6. Quality Attributes & Acceptance Tests

The system will be accepted when all high-priority functional requirements are implemented and verified, no critical security or data-integrity failures remain, required NFR thresholds are met, and the RTM demonstrates traceability from requirements to test cases.

Quality Attribute	Acceptance Criterion
Functional correctness	All high-priority FRs pass their associated test cases.
Performance	CSQP-NF-001 threshold is met under the agreed test load.
Reliability	CSQP-NF-002 availability criterion is satisfied for the evaluation period.
Security	All security requirements pass and no critical authentication/ authorization vulnerabilities remain.
Usability	Core workflows—login, material discovery, material contribution, quiz attempt, and result viewing—are completed without blocking usability defects.
Data integrity	Committed material, versions, quizzes, attempts, and scores remain consistent across create/read/update/retrieve tests.

6.1 Acceptance Test Suite

Test Case	Area	Scenario	Expected Result
TC-AUTH-01	Authentication	Valid/invalid username and password	Valid user authenticates; invalid credentials rejected.
TC-MAT-01	Material Upload	Valid and invalid PDF/note uploads	Valid upload succeeds; invalid type/ size rejected.
TC-MAT-04	Material Search	Search by title/subject/topic	Matching material is returned accurately.
TC-COL-01	Collaboration	Two authorized contributors update shared material	Changes are stored and versioned.
TC-QUIZ-03	Quiz Attempt	Start, answer, and submit quiz	Attempt is stored and submission completes.
TC-QUIZ-04	Evaluation	Known answer set with expected score	Calculated score matches expected score.
TC-QUIZ-07	Leaderboard	Multiple completed attempts	Ranking follows configured criteria.
TC-ADMIN-01	Administration	Change account status/role	Subsequent authorization reflects change.
TC-SEC-04	Authorization	Direct protected-resource request by unauthorized role	Request is rejected.
TC-PERF-01	Performance	Representative normal-load test	90th percentile interaction response ≤ 3 seconds.

7. UML Use-Case Diagrams

The following diagrams identify the principal actors and major use cases. They are intended as high-level system models; detailed sequence and class diagrams may be produced during the design phase.

7.1 UML Use-Case Diagram — Study Material Management

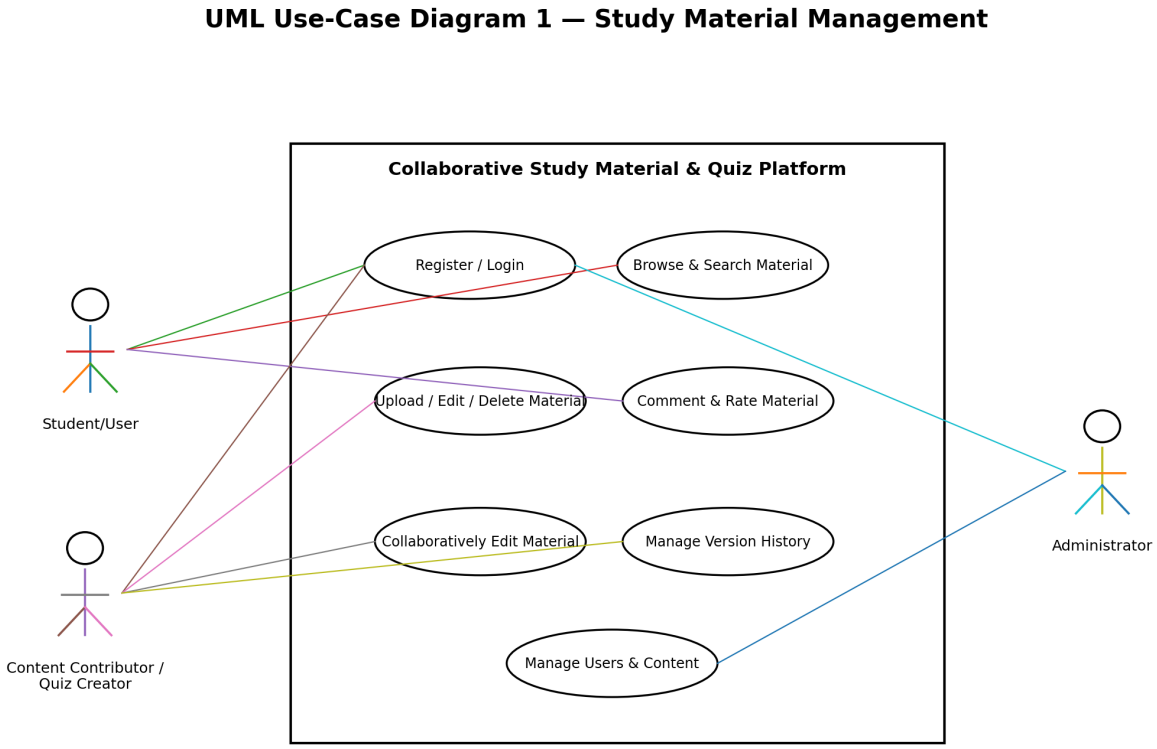


Figure 1. Major interactions for authentication, material discovery, contribution, collaboration, versioning, comments/ratings, and administration.

7.2 UML Use-Case Diagram — Quiz & Performance Management

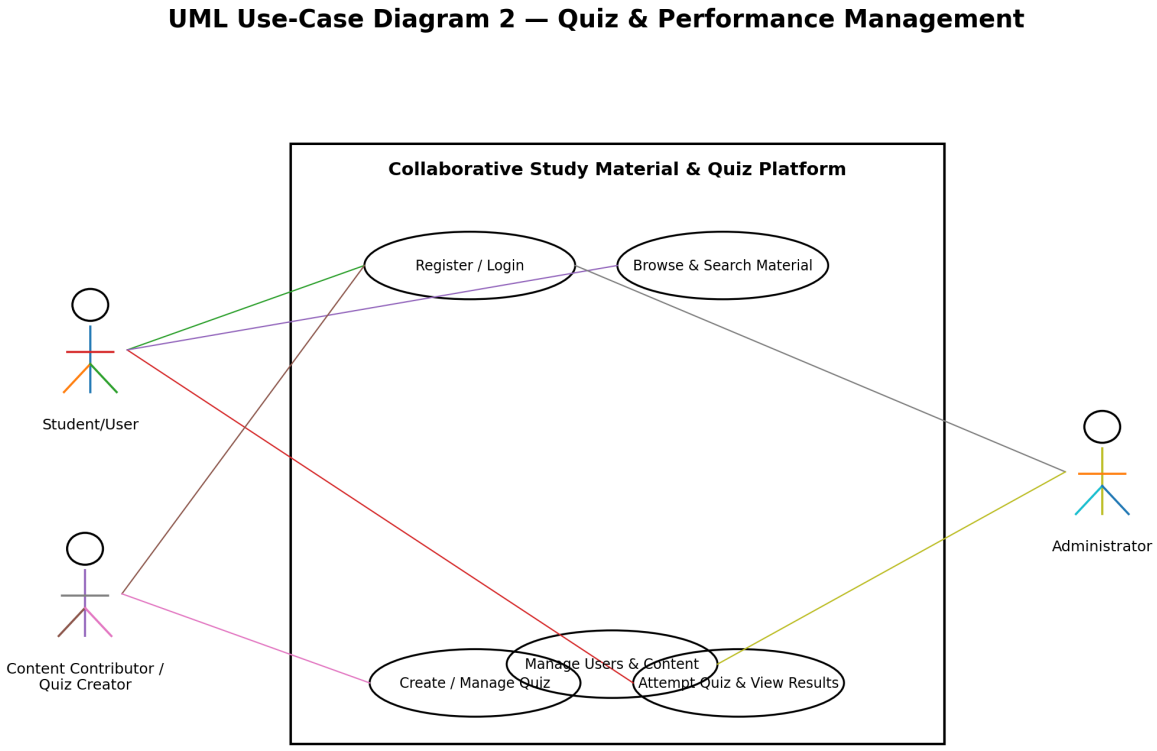


Figure 2. Major interactions for quiz creation, quiz attempts, result generation, material access, and administration.

8. Requirements Traceability Matrix (RTM)

The RTM connects each requirement to its implementation/design area and verification test. Status is marked N (Not yet implemented), P (Partially implemented), or A (Accepted) at the SRS stage.

Req ID	Requirement Short	Section	Module	Test Case(s)	Status (N/P/A)	Comments
CSQP-F-001	authenticate users using a unique username and password before granting access to protected functions	4	Authentication & RBAC	TC-AUTH-01	N	Design/implementation pending
CSQP-F-002	enforce role-based permissions for Student/User, Content Contributor / Quiz Creator, and Administrator roles	4	Authentication & RBAC	TC-AUTH-02	N	Design/implementation pending
CSQP-F-003	allow authorized contributors to upload study notes and PDF documents with configured file-size and file-type validation	4	Study Material	TC-MAT-01	N	Design/implementation pending
CSQP-F-004	allow authorized contributors to create study notes directly within the platform	4	Study Material	TC-MAT-02	N	Design/implementation pending

CSQP-F-005	organize study material by subject and topic and allow users to browse the hierarchy	4	Study Material	TC-MAT-03	N	Design/implementation pending
CSQP-F-006	provide search and filtering facilities for study material using supported metadata such as title, subject, topic, and contributor	4	Study Material	TC-MAT-04	N	Design/implementation pending
CSQP-F-007	allow users with appropriate permissions to download available study material	4	Study Material	TC-MAT-05	N	Design/implementation pending
CSQP-F-008	allow authorized contributors to edit or delete study material they are permitted to manage	4	Study Material	TC-MAT-06	N	Design/implementation pending
CSQP-F-009	allow users to rate study material using the configured rating scale and prevent duplicate ratings by the same user for the same item	4	Study Material	TC-MAT-07	N	Design/implementation pending
CSQP-F-010	allow users to post comments and participate in discussions associated with study material	4	Study Material	TC-MAT-08	N	Design/implementation pending
CSQP-F-011	maintain version history for collaboratively edited study material and allow authorized users to view previous versions	4	Collaboration & Versioning	TC-MAT-09	N	Design/implementation pending
CSQP-F-012	allow multiple authorized contributors to collaborate on shared study material through controlled edit/update operations	4	Collaboration & Versioning	TC-COL-01	N	Design/implementation pending
CSQP-F-013	allow authorized quiz creators to create quizzes containing multiple-choice, True/False, and fill-in-the-blank questions	4	Quiz Engine	TC-QUIZ-01	N	Design/implementation pending
CSQP-F-014	allow quiz creators to configure quiz duration and enable or disable question randomization	4	Quiz Engine	TC-QUIZ-02	N	Design/implementation pending
CSQP-F-015	allow eligible users to start, navigate, and submit a quiz attempt	4	Quiz Engine	TC-QUIZ-03	N	Design/implementation pending
CSQP-F-016	automatically evaluate objective quiz responses and calculate the user's score according to the quiz's configured marking rules	4	Quiz Engine	TC-QUIZ-04	N	Design/implementation pending
CSQP-F-017	display quiz results immediately after submission when the quiz is configured for instant results	4	Quiz Engine	TC-QUIZ-05	N	Design/implementation pending
CSQP-F-018	maintain a history of a user's completed quiz attempts, scores, and relevant timestamps	4	Quiz Engine	TC-QUIZ-06	N	Design/implementation pending

CSQP-F-019	provide a leaderboard for quizzes using the configured ranking criteria	4	Quiz Engine	TC-QUIZ-07	N	Design/implementation pending
CSQP-F-020	allow administrators to manage user accounts, including viewing, enabling/disabling, and applying role changes where permitted	4	Administration & Audit	TC-ADMIN-01	N	Design/implementation pending
CSQP-F-021	allow administrators to moderate study material, comments, and other user-generated content	4	Administration & Audit	TC-ADMIN-02	N	Design/implementation pending
CSQP-F-022	record important user and administrative actions in an audit log without storing plaintext passwords	4	Administration & Audit	TC-SEC-01	N	Design/implementation pending
CSQP-NF-001	For normal load, 90% of standard page/API interactions shall complete withi...	5	Performance	TC-PERF-01	N	NFR verification pending
CSQP-NF-002	The platform shall achieve at least 99.0% monthly availability excluding pl...	5	Reliability	TC-REL-01	N	NFR verification pending
CSQP-NF-003	The user interface shall provide consistent navigation, readable feedback, ...	5	Usability	TC-UX-01	N	NFR verification pending
CSQP-NF-004	The system shall preserve data integrity so that saved material, quiz defin...	5	Data Integrity	TC-DATA-01	N	NFR verification pending
CSQP-NF-005	The platform shall support maintainable modular components and documented i...	5	Maintainability	TC-MAINT-01	N	NFR verification pending
CSQP-NF-006	The platform shall support the configured study-material and quiz workload ...	5	Scalability	TC-SCALE-01	N	NFR verification pending
CSQP-SR-001	The system shall store passwords only as secure one-way password hashes usi...	5.2	Security	TC-SEC-02	N	Security verification pending
CSQP-SR-002	The system shall use HTTPS/TLS for authenticated sessions and other sensiti...	5.2	Security	TC-SEC-03	N	Security verification pending
CSQP-SR-003	The system shall enforce server-side authorization checks for all protected...	5.2	Security	TC-SEC-04	N	Security verification pending
CSQP-SR-004	The system shall validate and sanitize user-provided content and uploaded-f...	5.2	Security	TC-SEC-05	N	Security verification pending
CSQP-SR-005	The system shall terminate or invalidate authenticated sessions according t...	5.2	Security	TC-SEC-06	N	Security verification pending
CSQP-SR-006	The system shall prevent users from altering quiz scores or other authorita...	5.2	Security	TC-SEC-07	N	Security verification pending

Appendix A — Requirement Summary

This SRS contains 22 functional requirements, 6 non-functional requirements, 2 security objectives, and 6 security requirements. The counts exceed the minimum requirements specified in the provided reference SRS.